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Page 10 of 10

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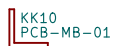


KK6  
SP400-0.007-00-104

□

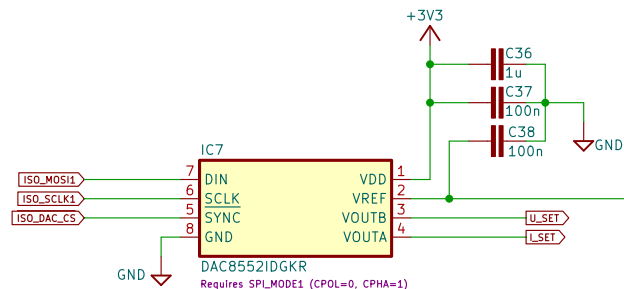
 KK7  
M3X8/D7985B

Id: 1/8

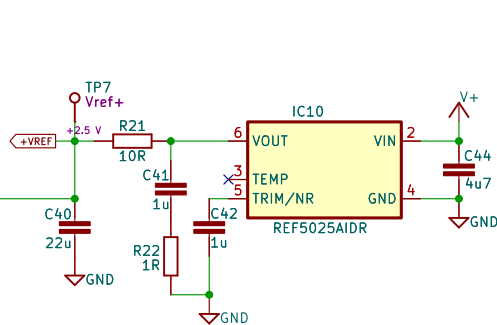




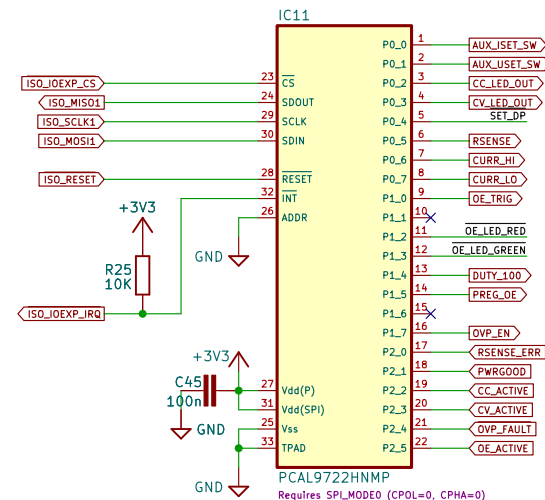
# 16-bit DAC (set voltage and current)



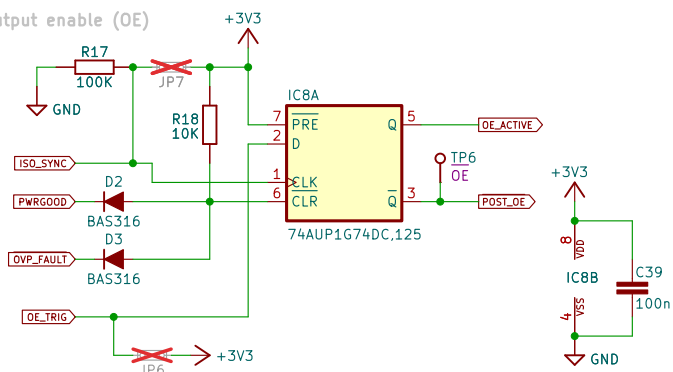
# Voltage reference



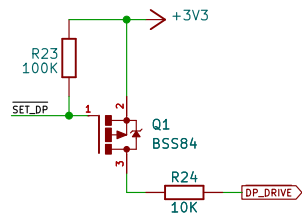
# SPI 22-bit I/O Expander



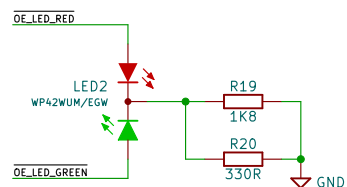
# Output enable (OE)



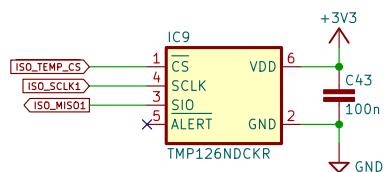
# Down-programmer



# OE indicator



# SPI Temperature sensor



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Sheet: /DAC, OE, Vref, IOEXP, Temp/

File: DAC\_Vref\_IOEXP\_Temp.kicad\_sch

**Title: EEZ DIB DCP405+ power module**

Size: A4

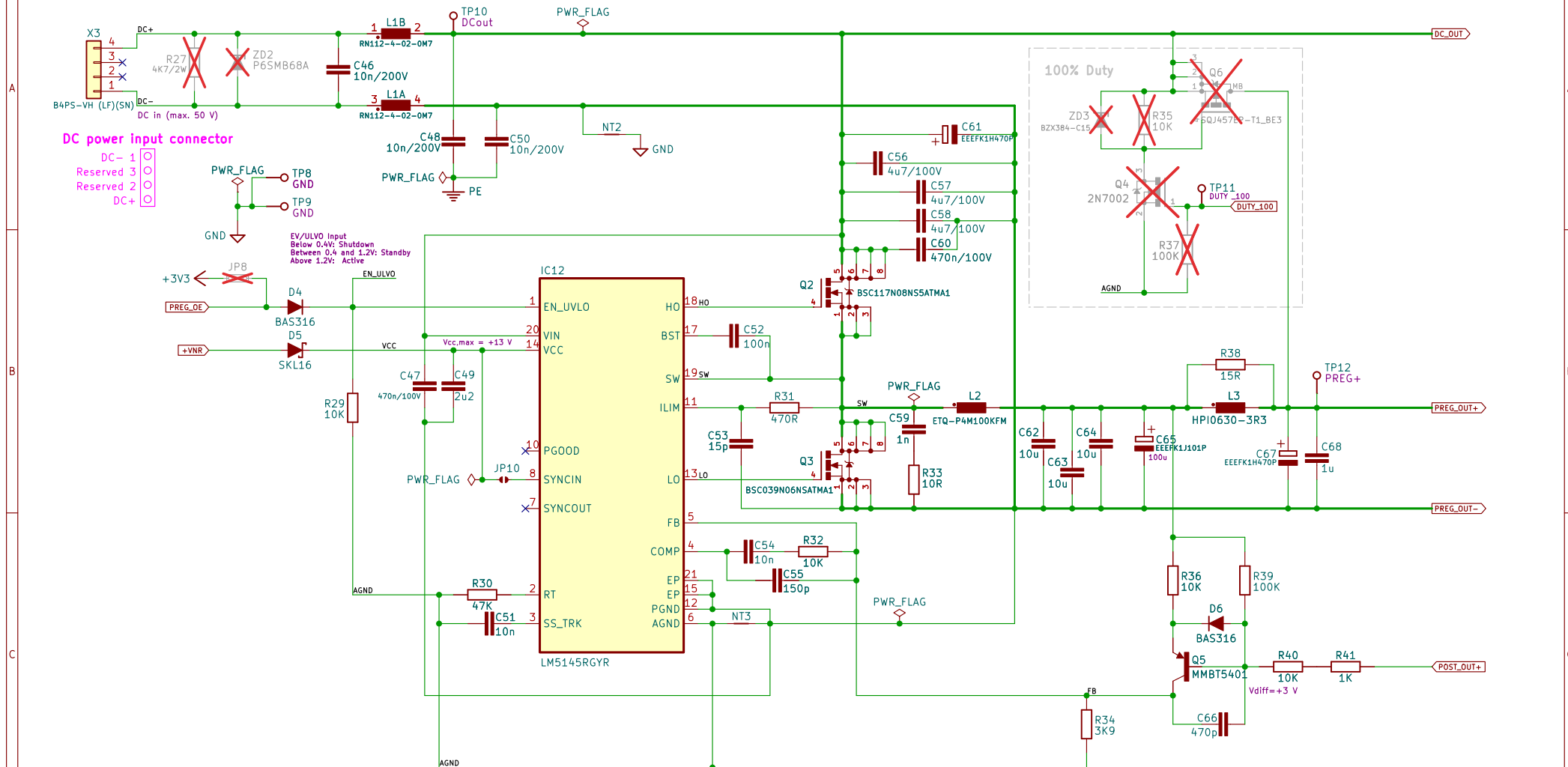
Date: 2026-09-21

Rev: r4B5

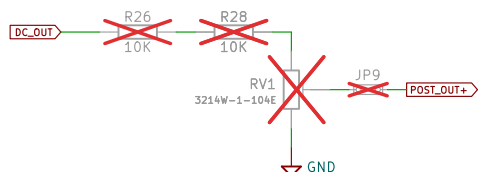
KiCad E.D.A. 10.0.6

Id: 3/8

# Low EMI power pre-regulator



## Test point



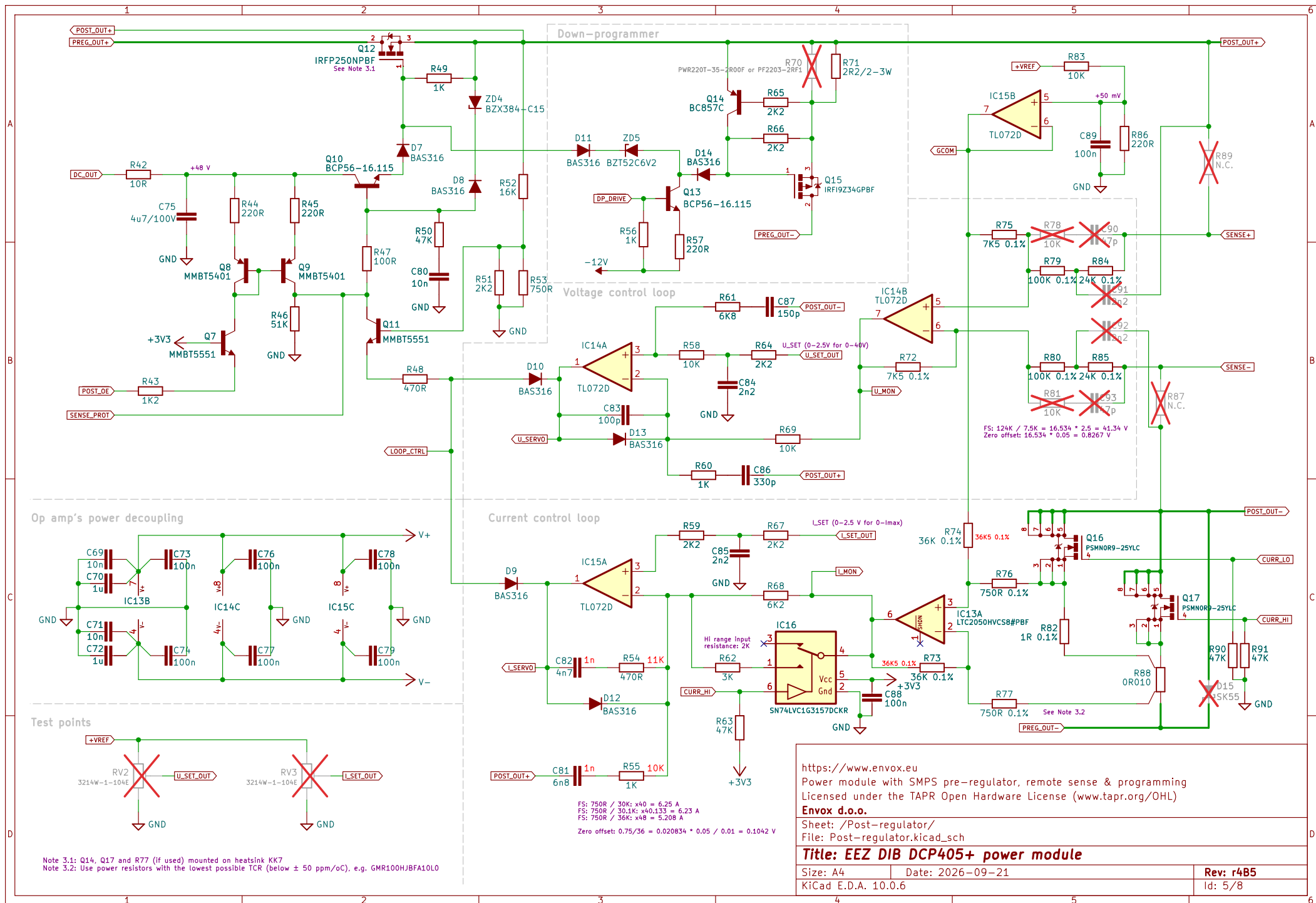
<https://www.envox.eu>  
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**Envox d.o.o.**

Sheet: /Power pre-regulator/  
 File: Power\_pre-regulator.kicad\_sch

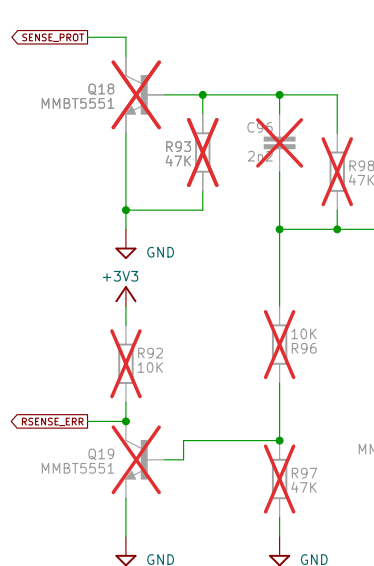
## Title: EEZ DIB DCP405+ power module

Size: A4 Date: 2026-09-21  
 KiCad E.D.A. 10.0.6

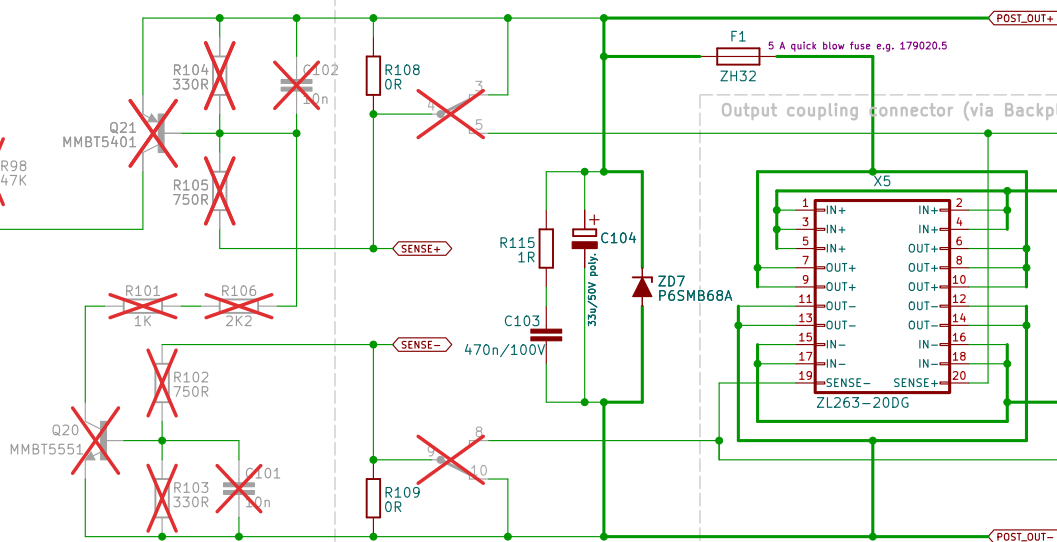
Rev: r4B5  
 Id: 4/8



## Remote sense reverse polarity detection



## Power output section

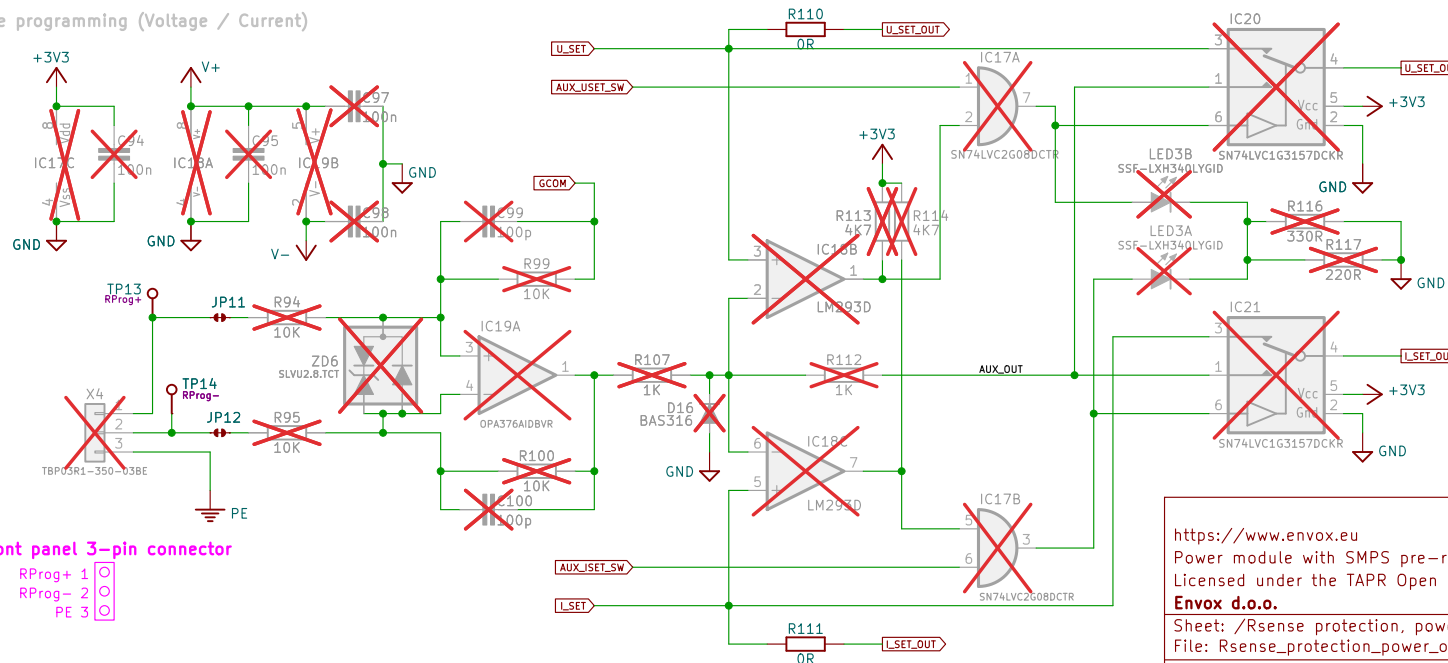


Output coupling connector (via Backplane PCB)

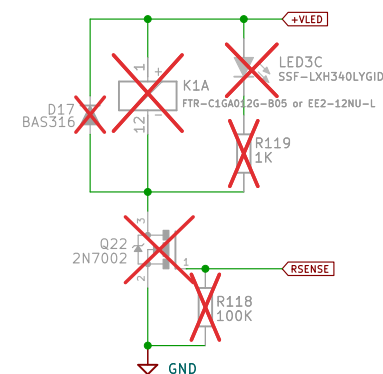
20-pin DIB 2.0 power socket

|              |              |
|--------------|--------------|
| OUTC+ 1      | 2 OUTC+      |
| OUTC+ 3      | 4 OUTC+      |
| OUTC+ 5      | 6 POST_OUT+  |
| POST_OUT+ 7  | 8 POST_OUT+  |
| POST_OUT+ 9  | 10 POST_OUT+ |
| POST_OUT+ 11 | 12 POST_OUT+ |
| POST_OUT+ 13 | 14 POST_OUT+ |
| OUTC- 15     | 16 OUTC-     |
| OUTC- 17     | 18 OUTC-     |
| SENSE- 19    | 20 SENSE+    |

## Remote programming (Voltage / Current)



## Remote sense selection



## Front panel 3-pin connector

|          |   |
|----------|---|
| RProg+ 1 | 2 |
| RProg- 2 | 3 |
| PE 3     | 4 |

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Envox d.o.o.

Sheet: /Rsense protection, power output, Rprog/

File: Rsense\_protection\_power\_output.kicad\_sch

**Title: EEZ DIB DCP405+ power module**

Size: A4

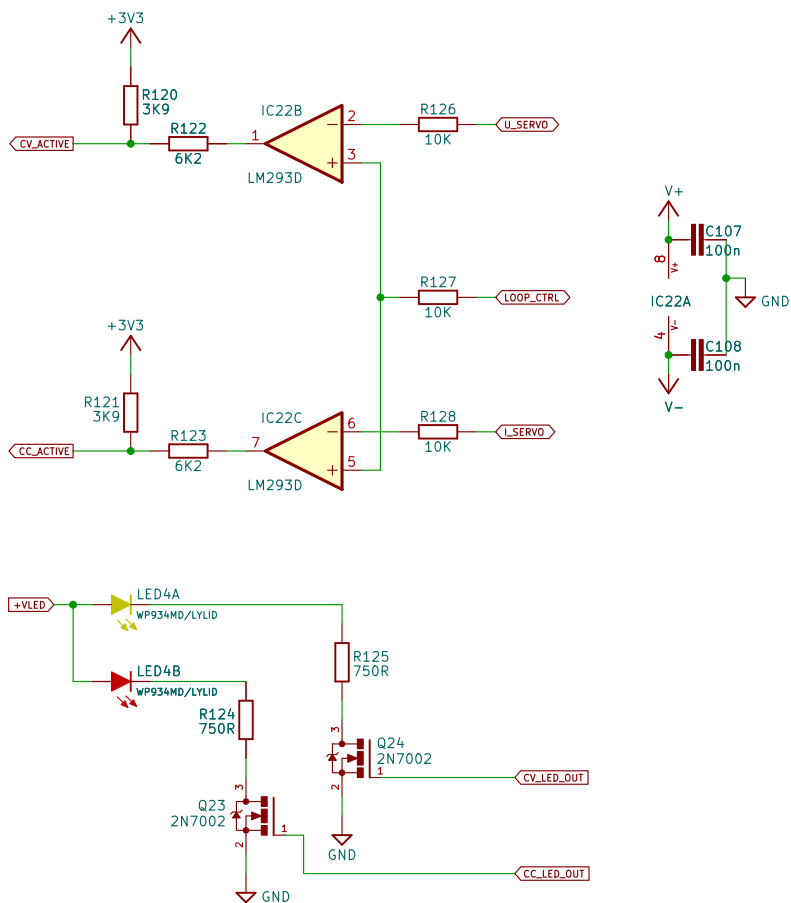
Date: 2026-09-21

Rev: r4B5

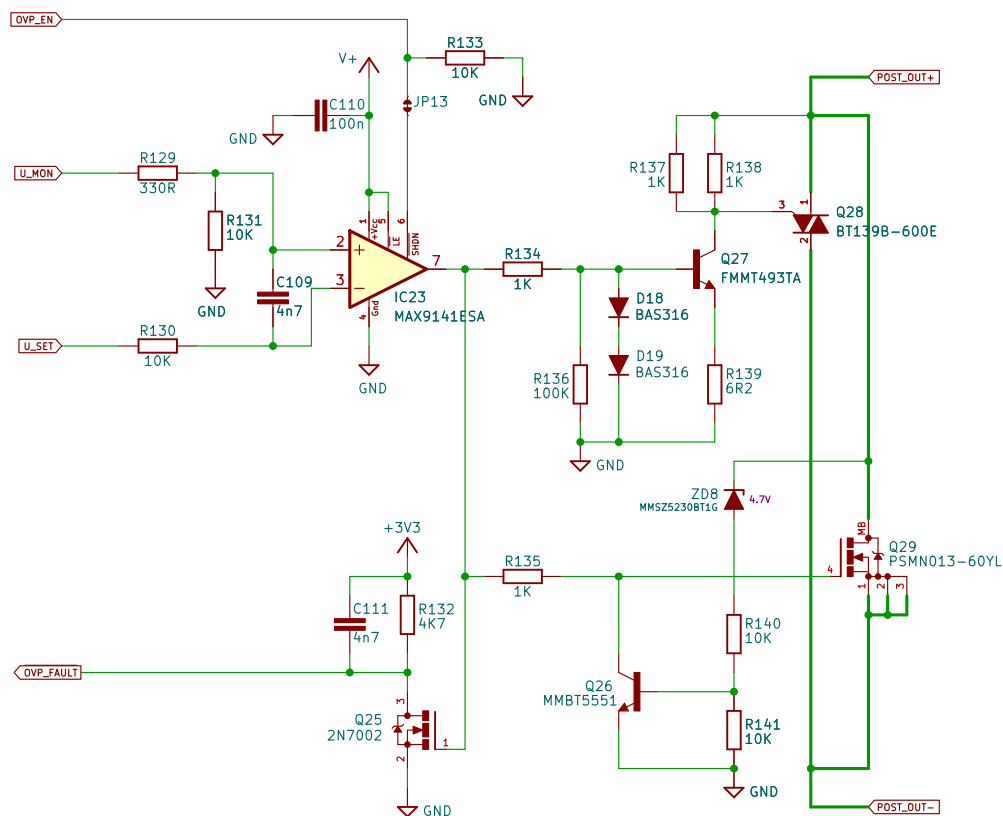
KiCad E.D.A. 10.0.6

Id: 6/8

# Constant Voltage (CV) and Constant Current (CC) mode detection and indication



# Over-voltage protection (OVP)



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**Envox d.o.o.**

Sheet: /Mode detection, OVP/

File: CC\_CV\_OVP.kicad\_sch

**Title: EEZ DIB DCP405+ power module**

Size: A4

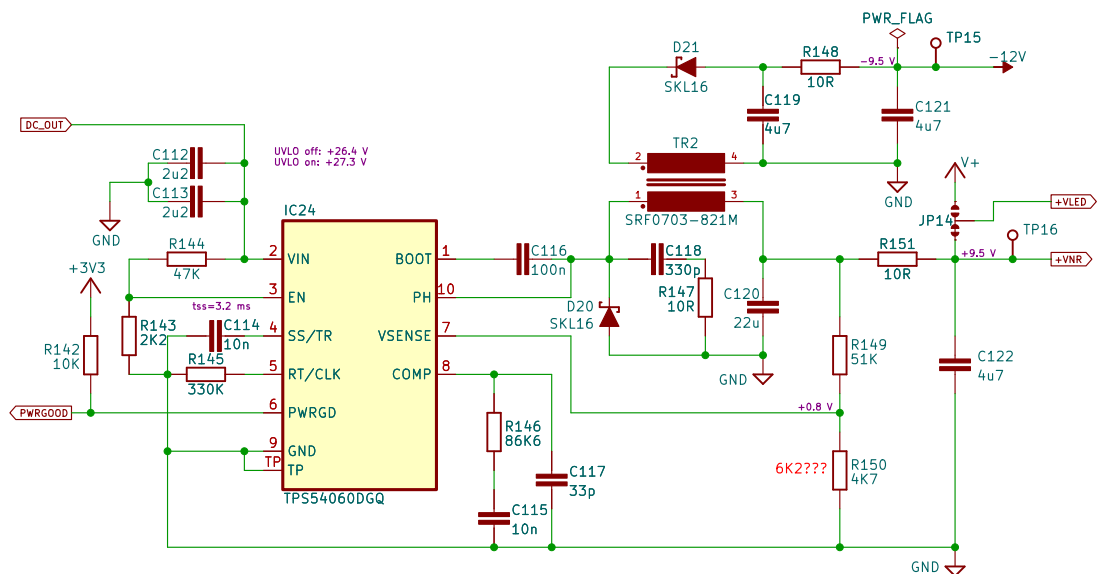
Date: 2026-09-21

Rev: r4B5

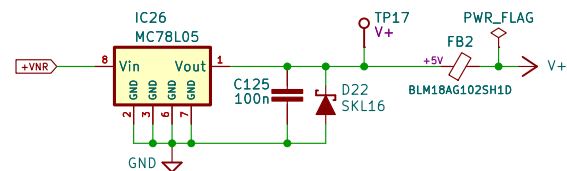
KiCad E.D.A. 10.0.6

Id: 7/8

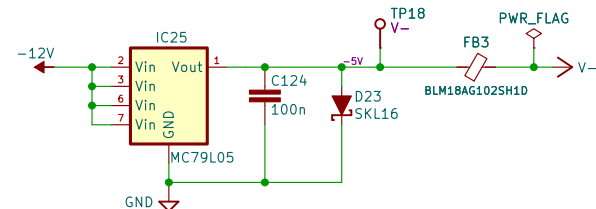
# Bias power supply



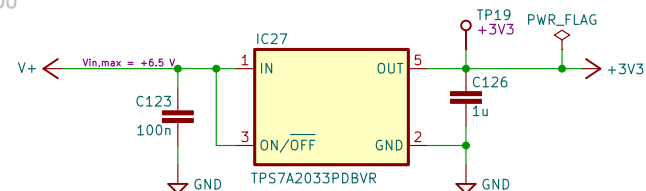
## +5V LDO



## -5V LDO



## +3V3 LDO



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**Envox d.o.o.**

Sheet: /Bias power/

File: Bias\_power.kicad\_sch

**Title: EEZ DIB DCP405+ power module**

Size: A4

Date: 2026-09-21

Rev: r4B5

KiCad E.D.A. 10.0.6

Id: 8/8